PROJECT PROPOSAL

Low-Cost LoRa Beacon System for Elephant Proximity Alerts: RF Front-End Design

Request for access to a VNA and two USRPs: SPECTRA 2026 RF Circuit Design Challenge

Field	Details
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1. Summary

Wild elephants regularly enter villages around Anuradhapura and Polonnaruwa, damaging crops and endangering lives. We propose a low-cost early-warning system: a LoRa beacon on the elephant collar and receivers placed around the village that trigger a siren when an elephant comes near. Our competition entry focuses on the RF front-end of the board (impedance matching, harmonic filtering and TX/RX switching) built around an STM32WL microcontroller with an integrated LoRa radio. To verify the design, we request supervised access to one vector network analyzer (VNA) and two USRPs for [number] hours per week over [duration].

2. Background and Problem

- Human–elephant conflict is a serious problem near Anuradhapura and Polonnaruwa.
- Existing GPS collars are expensive. For example, the Smart Parks ElephantEdge tracker is priced at about €699, which is out of reach for village-level deployment.
- Villages lack a simple, affordable early-warning system.

3. Competition Context

- Competition: SPECTRA 2026, RF Circuit & Antenna Design Competition, organized by the University of Moratuwa IEEE MTT-S Student Chapter.
- Our category: RF Circuit Design Challenge.
- Key requirements: operation strictly below 3 GHz; input match S11 ≤ −10 dB at the design frequency; simulation, a physical PCB prototype and tolerance (variance) analysis; live measurement by a judging panel; RF power at the test port limited to 0 dBm.
- Stages: (1) Design Proposal Screening, (2) Finals. Dates will be announced by the organizers.

4. Objectives

- Design and fabricate an RF front-end in the 433/868 MHz region with S11 ≤ −10 dB at the declared design frequency.
- Measure and report harmonic suppression and transmit power.
- Perform a component-tolerance (variance) analysis and identify the most sensitive components.
- Measure path loss in the field and demonstrate RSSI-based proximity detection.

5. Proposed System

Collar beacon (STM32WL + RF front-end) transmits a short LoRa packet periodically → village receivers measure signal strength → when the strength indicates an elephant is near, a siren is triggered.

- **Competition scope:** the RF front-end (matching network, low-pass harmonic filter, TX/RX switch and, on the receiver, an optional band-pass filter and LNA) ending in a 50 Ω SMA port.
- **Supporting demonstration:** the STM32 core, I2C expansion ports and siren logic.

6. Technical Plan

- **Phase 1, Simulation:** matching network and filter simulation, S-parameters and Monte Carlo tolerance analysis.
- **Phase 2, Fabrication:** design and build the PCB (50 Ω microstrip on FR4, SMA test port).
- **Phase 3, Bench measurement:** S11/S21 with a VNA, spectrum and harmonic checks with a USRP, comparison against simulation.
- **Phase 4, Field test:** RSSI and path-loss measurement, proximity-detection logic, siren demonstration.

7. Equipment Request and Justification

Equipment	Intended use	Why it is needed
VNA (1)	Measure S11 and S21 of the matching network, filter and switch paths before the competition.	Calibrated, higher-accuracy measurements let us correct the design before the judges measure it on a nanoVNA.
USRP (2)	One as a spectrum analyzer to verify transmit power and harmonic levels; the second as a reference receiver to capture LoRa packets over a cable; log signal strength for path-loss characterization.	The available RTL-SDR is receive-only and cannot generate RF signals for controlled testing. A USRP provides both transmission and reception, enabling spectrum, harmonic, and signal-quality verification with greater flexibility and measurement capability.

8. Usage Plan, Safety and Compliance

- All transmitter tests will be cabled through attenuators. A transmitter will never be connected directly to a USRP or VNA input.
- Any over-the-air transmission will take place only with prior approval of the competition organizers and in line with applicable Sri Lankan spectrum regulations.
- The team accepts responsibility for the equipment, will make no modifications and will return it in its original condition.

9. Timeline

Period	Activity	Equipment needed
Weeks 1–3	Simulation and tolerance analysis; Stage 1 proposal	VNA, USRPs
Weeks 4–9	PCB design and fabrication, Bench measurement and design correction	VNA, USRPs
Weeks 10–11	Field test and proximity logic	VNA, USRPs
Weeks 12+	Report, presentation and Final's preparation	VNA, USRPs

10. Resources and Cost

- The team will fund the PCBs, components and STM32 boards.

11. Expected Outcomes and Benefits

- Representation of the department in a national IEEE student competition.
- A project addressing a real social and environmental problem in Sri Lanka.
- A fabricated, measured RF front-end with a full technical report.
- A reusable measurement dataset and demonstration.